Surveillance, case investigation and contact tracing in filovirus disease outbreaks

AfricaCDC
Centres for Disease Control and Prevention

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Abbreviations

CFR	case fatality rate
EVD	Ebola virus disease (caused by Ebola virus, species *Orthoebolavirus zairense*)
HCW	health-care worker
IDSR	integrated disease surveillance and response
IHR	International Health Regulations (2005)
IPC	infection prevention and control
MVD	Marburg virus disease (caused by Marburg or Ravn virus, species *Orthomarburgvirus marburgense*)
NFP	national IHR focal point
PoE	point of entry
RNA	ribonucleic acid
WHO	World Health Organization

Key points

- The primary goals of filovirus disease surveillance, case investigation and contact tracing are to rapidly detect outbreaks, detect all cases, interrupt chains of transmission and guide the overall operational response. This aims to protect communities at risk in affected areas, reduce morbidity and mortality and ultimately stop human-to-human transmission of filoviruses.
- Routine surveillance with early warning capacity should be established, case and contact person definitions adapted, and flows for case isolation and laboratory confirmation identified.
- A dedicated alert system should be set up to identify and isolate all potential new suspected cases as soon as possible after symptom onset.
- The threshold for action is one suspected case.
- As soon as a suspected case is identified, isolation and laboratory confirmation should be undertaken, as well as case investigation, including identification of possible sources of exposure and identification of contact persons during the infectious period, should the suspected case be subsequently classified as probable or confirmed.
- Contact persons of probable and confirmed cases of filovirus disease should be listed and followed up for 21 days.
- Identifying, notifying and following up contact persons across international borders (if applicable) is an integral component of outbreak control and must be planned from the outset.
- Epidemiological surveillance of all morbidities and mortalities should be maintained during filovirus disease outbreaks, notably for vaccine-preventable diseases and malaria.
- All surveillance activities that involve interaction with the community should be supported and coordinated with risk communication and community engagement teams.
- All data collected should be analysed with a defined purpose and findings shared with all relevant stakeholders, including the affected communities.
- Surveillance data should be analysed and interpreted considering the context and in close collaboration with all involved stakeholders.
- Data should be anonymized to the greatest possible extent and should be protected.

1. Introduction

1.1 Background

This document provides operational guidance for public health professionals engaged in surveillance and outbreak response to filovirus disease outbreaks, such as Ebola disease, caused by Bundibugyo virus, Ebola virus, Sudan virus and Taï Forest virus, and Marburg virus disease (MVD), caused by Marburg or Ravn virus (see Box 1).

This guidance is an update of the September 2015 emergency guideline on *Implementation and management of contact tracing for Ebola virus disease: emergency guideline* (*1*) and covers the full surveillance and epidemiological cycle from routine preparedness to active outbreak investigation and analytics. It is structured around five operational functions:

- routine surveillance in humans and wildlife
- early warning and alert systems
- outbreak investigation and contact tracing
- epidemiological analytics
- community engagement throughout all phases.

This document should be read alongside existing World Health Organization (WHO) technical documents related to filovirus diseases, in particular the *Early warning alert and response in emergencies: an operational guide* (*2*), the *WHO guideline on contact tracing* (*3*), the *Diagnostic testing for Ebola disease and Marburg virus disease: interim guidance, 9 July 2026* (*4*), the *Infection prevention and control guideline for Ebola and Marburg diseases* (*5*) and the *Integrated outbreak analytics toolkit* (*6*). It does not replace national protocols, which should be developed in accordance with national legislation and the International Health Regulations (2005) (IHR) (*7*).

Box 1. Key filoviruses covered by this guidance (as of September 2026)

Ebola virus (species *Orthoebolavirus zairense*), causing Ebola virus disease (EVD)

Case fatality rate (CFR) of up to 90%. This species has caused the largest recorded outbreaks, including the 2013–2016 West Africa outbreak. A licensed vaccine and therapeutics are available, and a ring vaccination strategy is applicable.

Bundibugyo virus (species *Orthoebolavirus bundibugyoense*), causing Bundibugyo virus disease

The CFR is approximately 25–36%. It was first identified in 2007 in Uganda. No vaccines or therapeutics have been licensed; however the *WHO emergency guidance on the use of the Ebola virus vaccine during Bundibugyo virus outbreaks, 28 May 2026*, is available for potential emergency use (*8*). Candidate therapeutics and vaccines are under evaluation.

Box 1. Key filoviruses covered by this guidance (as of September 2026) *(cont.)*

Sudan virus (species *Orthoebolavirus sudanense*), causing Sudan virus disease

The CFR is approximately 41–65%. Multiple outbreaks have occurred in Uganda and South Sudan. No vaccines or therapeutics have been licensed as of September 2026; candidate therapeutics and vaccines are under evaluation.

Marburg virus and Ravn virus (species *Orthomarburgvirus marburgense*), causing Marburg virus disease (MVD)

The CFR has been approximately 24–88% in past outbreaks. No vaccines or therapeutics have been licensed; candidate therapeutics and vaccines are under evaluation.

1.2 Target audience

This guidance is intended for national and subnational public health practitioners, epidemiologists, surveillance officers, laboratory staff and field responders involved in the detection, investigation and response to filovirus outbreaks.

1.3 Methodology

This guidance builds on the 2015 *Implementation and management of contact tracing for Ebola virus disease: emergency guideline* (*1*), that was developed by WHO in collaboration with technical experts, partners and ministries of health, and incorporates current available evidence on filoviruses, and from other WHO normative guidance and technical documents on implementing surveillance systems during disease outbreaks, including the *Operational guide for engaging communities in contact tracing, 28 May 2021* (*9*), the 2023 *Early warning alert and response in emergencies: an operational guide* (*2*) and the 2025 *WHO guideline on contact tracing* (*3*).

The filovirus disease case definitions for routine surveillance and for filovirus disease outbreak contexts, initially developed in 2015 for the Ebola virus disease outbreak that affected West Africa were reviewed based on more recent evidence from outbreaks which occurred in the past eleven years and with the input and guidance of technical experts. The evidence was collected from suspected, confirmed and probable filovirus disease case descriptions reported in the past eleven years. Discussions were organized and consensus achieved during two dedicated meetings hosted by WHO and the Africa Centres for Disease Control and Prevention.

The guidance is unique in the way it explores each step to implement a functional surveillance system and elaborates on epidemiological analyses needed to efficiently monitor a filovirus disease outbreak and inform response interventions. It was reviewed by a group of technical experts with extensive filovirus disease and field epidemiology experience, listed in the acknowledgement section.

2. Natural history and epidemiology of filovirus diseases

2.1 Virus characteristics

Filoviruses are single-stranded, negative-sense ribonucleic acid (RNA) viruses with characteristic filamentous morphology. They are classified in the family *Filoviridae* and are among the most pathogenic known viruses in humans, with CFR ranging from approximately 25–90% in untreated populations, depending on the virus species, outbreak context and access to supportive or targeted care (see Box 2).

Box 2. Key epidemiological features of filovirus diseases

Incubation period: 2–21 days

Infectious period: From symptom onset; infectivity increases with disease severity and is highest in the late stages and at death.

Clinical presentation: The symptoms can be sudden and include fever, fatigue, malaise, muscle pain, headache and sore throat. These are followed by vomiting, diarrhoea, abdominal pain, rash and symptoms of impaired kidney and liver functions.

Exposures: Direct physical contact with infected animals or with body fluids or remains of confirmed and probable cases (for example, among health-care workers (HCW), family caregivers or funeral attendees), particularly when appropriate infection prevention and control (IPC) measures are not available or adequately used.

2.2 Animal reservoir

Fruit bats (Pteropodidae family) are considered the most likely natural reservoir for orthoebolaviruses, based on seroprevalence studies and limited detection of viral RNA and antibodies in several bat species, particularly *Hypsignathus monstrosus*, *Epomops franqueti* and *Myonycteris torquata*. *Rousettus aegyptiacus* has been identified as the natural reservoir for orthomarburgviruses, with multiple outbreak investigations confirming direct bat-to-human transmission in cave and mine settings *(10)*.

Despite decades of research, complete isolation of infectious *Orthoebolavirus* from bats has proved elusive, underscoring the importance of continued ecological and virological research.

The investigation of bats in the field of emerging zoonoses requires balancing ecological, conservation and public health interests, as detailed in the Food and Agriculture Organization of the United Nations' guidance on *Investigating the role of bats in emerging zoonoses: balancing ecology, conservation and public health interests* (*11*).

2.3 Transmission: animals to humans (spillover events)

Human outbreaks of Ebola disease and MVD typically begin with a zoonotic spillover event – the introduction of the virus from an infected animal into a human host. Documented routes of zoonotic transmission include:

- direct contact with infected fruit bats (hunting, handling, consumption of bat meat)
- direct contact with non-human primates (gorillas, chimpanzees) or other wild mammals (duikers) that have themselves been infected through exposure to bats or their excretions
- exposure (prolonged stay) in mines or caves inhabited by *Rousettus* bats (*Orthomarburgvirus*).

The index case in an outbreak is usually a person living or working in or near a forest with frequent exposure to forest environments or wildlife. In five outbreaks reported in the Congo and Gabon between October 2001 and December 2003, four of five EVD outbreaks were associated with exposure of index human case patients to infected animals (*13*). Similar sources have been documented for MVD (*14*).

Identifying the index case and the likely source of spillover is a primary objective of outbreak investigations. It should (i) help to control the source, as needed, if multiple spillovers are occurring; and (ii) help inform the ecology of the disease and potential mitigation measures for future outbreaks.

2.4 Transmission: human-to-human

Human-to-human transmission requires contact (direct or indirect) with the blood, secretions, other body fluids or organs of an infected and symptomatic (or deceased) person. There is no documented evidence of human transmission during the pre-symptomatic phase of the disease.

Transmission occurs through two means:

- direct contact through broken skin or mucous membranes with the blood or body fluids of a person who is sick with or has died from filovirus disease; or
- indirect contact with broken skin or mucous membranes through surfaces or objects contaminated with body fluids of sick persons or persons who have died.

Filoviruses do not normally spread through respiratory droplets or the airborne route.

Examples of exposures may include:

- needlesticks/sharps injuries with contaminated equipment
- contact during funerals, particularly when preparing a recently deceased body for burial
- close skin-to-skin contact between sick mother and child

Health care-associated transmission and amplification is a major risk in settings without adequate IPC measures.

2.5 Viral persistence after recovery and potential secondary transmissions

Ebola virus, Marburg virus and Sudan virus RNA have been detected in immune-privileged sites (for example, eyes, placenta and testicles) and body fluids, including semen, breast milk in women who were lactating during acute illness and cerebrospinal fluid, and it can persist for prolonged periods following clinical recovery in some survivors. Transmission linked to infected semen from male survivors, or relapse in individuals who have recovered, has been documented in rare instances and can lead to secondary transmission and outbreaks, sometimes months after the end of a declared outbreak.

Survivors require dedicated follow-up, including counselling, clinical and psychosocial care, biological testing and other services to help them cope with sequelae of the disease (*15*).

3. Surveillance and outbreak investigation

3.1 Establishing early warning capacity through integrated disease surveillance

Early warning, alert and response aims to identify all suspected cases as early as possible, including before sustained transmission is established, ensure timely laboratory confirmation and trigger a rapid, proportionate response. A well-designed surveillance system links and coordinates human health, animal health and environmental surveillance, ideally within a One Health framework.

3.1.1 Integrated disease surveillance

Filovirus diseases should be addressed within national integrated disease surveillance and response (IDSR) frameworks, as described in the 2019 WHO *Integrated disease surveillance and response technical guidelines* (*16, 17)*. IDSR provides the foundational architecture for detecting, reporting and responding to priority diseases, including those with epidemic potential.

In routine surveillance, suspected filovirus disease events may first be identified through signals reported by the community as part of event-based surveillance (Table 1). These signals should be reported immediately to the health-care facility and the surveillance officer in charge, then investigated using the health-care facility and surveillance case definition, together with local context and knowledge of filovirus ecology. Suspected cases identified at the health-care facility level tend to present later in the course of illness, and their detection should raise suspicion that additional cases are occurring in the community.

Once a case is laboratory-confirmed (*4*), it should be classified and reported as the specific filovirus disease (for example, EVD or MVD), in accordance with national IDSR guidance. A single confirmed human case of filovirus disease is sufficient to trigger the declaration of an outbreak in the affected locality.

The following components of surveillance should be operational, with appropriate geographical and population coverage prior to an outbreak to allow for early detection and investigation of filovirus outbreaks (see section 2.2.1 Preparedness checklist 1: Surveillance system in *Early warning alert and response in emergencies: an operational guide* (*2*)).

- Event-based surveillance systems monitor media, community rumours and other nontraditional sources, mostly at central level. Guidance to establish event-based surveillance is available from modules 5, 7 and 9 of *Early warning alert and response in emergencies: an operational guide* (*2*) and Annex 1C in the *Integrated disease surveillance and response technical guidelines, booklet two: sections 1, 2 and 3, third edition* (*17*).

- Indicator-based surveillance systems rely on the regular reporting of data, for example, by public and private health-care facilities, laboratories, the national health information system, health-care services monitoring, syndromic surveillance, mortality surveillance, veterinary services and wildlife services. Guidance to establish indicator-based surveillance is available from modules 5, 6 and 9 of *Early warning alert and response in emergencies: an operational guide* (*2*).

- Community indicator- and event-based surveillance systems use specific criteria that enable community health workers and community members to collect regular information on syndromes and report unusual illness or death events to the designated HCW or surveillance officer. Guidance to establish community-based surveillance is available from Annex E in the *Integrated disease surveillance and response technical guidelines, booklet one: introduction section, third edition* (*16*).

- Cross-border surveillance and collaboration mechanisms between neighbouring countries allow real-time, case-based epidemiological, laboratory and contact tracing data and information sharing. Guidance to establish cross-border surveillance is available from the interim guidance documents *Considerations for border health and points of entry for filovirus disease outbreaks* (*18*) and *Considerations for strengthening international information sharing for tracing and managing infectious disease cases and contact persons: interim guidance* (*19*) and *Bundibugyo Ebola virus disease: points of entry enhanced surveillance, interim guidelines, 9 June 2025* (*20*).

- Trained and equipped community health workers, HCWs, surveillance officers and epidemiologists can detect and verify signals, characterize alerts and investigate outbreaks.

- Established and identified communication and data flows, regular data review and feedback loops pass from national to subnational levels and between sectors.

HCWs are disproportionately affected during filovirus disease outbreaks, especially early in the outbreak, due to frequent contact with symptomatic patients in clinical settings, often before a diagnosis is confirmed. Risks are amplified when IPC measures are inadequate, facilities are overcrowded or if triage protocols are not in place.

Standardized case definitions for event-based surveillance that encompasses filovirus events and standardized suspected, probable and confirmed case definitions for filovirus disease should be disseminated to all levels of the health system, including to the community health workers (see Table 1).

Table 1. Filovirus case definitions to be used outside of a filovirus disease outbreak by detecting entity

Filovirus disease case definition to be used by health-care professionals	**Suspected case** a. Any individual presenting with fever or history of fever and no response to treatment of usual causes of fever in the area, **AND** • haemorrhage, including bloody diarrhoea, bleeding from gums, bleeding into skin (purpura), bleeding into eyes or urine; **OR** • at least two of the following symptoms: severe headache, weakness, joint pain, muscle pain, diarrhoea, vomiting, nausea, conjunctivitis, hiccups, irritability/fussiness (in children under the age of 2) **AND** recent history of visiting mines/ caves; exposure to sick or dead wild animals; being in contact with sick individuals with similar symptoms. **OR** b. Any individual who died after bleeding.
	Confirmed case a. A suspected case with laboratory confirmation[1]
Examples of events to be reported through community-based surveillance (not exhaustive)	• Two or more persons presenting with similar severe illnesses in the same setting within two weeks. • Two or more persons dying in the same community within two weeks. • Any human illness or death after exposure to animals and raw animal products (such as fluids and raw meat). • Cluster of illness or nontraumatic death from human or wild animal (likely to be reported by a health-care facility). • Cluster of illness or deaths among HCWs after attending to a similarly ill patient.

The detection of a suspected event or case of filovirus through any of the above-listed systems should trigger mandatory immediate reporting from the detecting entity to the health-care facility and/or the district, and from the district to the national level. Typically, any suspected case or event (such as a cluster of sickness or deaths among HCWs) detected by the community or a health-care facility should be reported to the district (or equivalent) and from district to national level within 24 hours of detection.

Countries, territories and areas that are at higher risk of importation of filovirus disease cases from a country, territory or area with a declared outbreak should use case definitions described in Table 2.

Given the rarity of filovirus disease, its high CFR, person-to-person transmission potential and the critical importance of speed in such responses, there is no minimum cluster threshold required before investigation is initiated. Box 3 describes key steps to undergo as soon as a suspected case is detected, and does not replace existing national guidance.

[1] The preferred assay to confirm a filovirus infection is nucleic acid amplification test; in certain circumstances genomic sequencing or serology (convalescent patients) may be used.

Box 3. One suspected case = Immediate action required

Upon identification of a single suspected case (alive or deceased), do the following.

1. Immediately isolate the suspected case, applying appropriate IPC precautions. Refer to clinical management if the individual is alive, and secure the body if the individual is dead (*5*).

2. Collect appropriate specimens and dispatch urgently to the designated laboratory, applying appropriate IPC precautions; specimen types for laboratory testing depend on the status of the patient (alive or deceased) (*4*).

3. Request testing for endemic circulating pathogens that may cause fever, such as malaria or typhoid, when relevant, applying appropriate IPC precautions.

4. Request testing for all filoviruses known to circulate or potentially present in the country.

5. Initiate listing for all contact persons of the suspected case-patient from symptom onset onwards, following the contact person definition criteria (Table 4), should the case-patient be subsequently classified as probable or confirmed. If the case-patient is classified as non-case, cease all contact tracing activities pertaining to filovirus disease outbreak management.

6. Notify the district and national health authorities within 24 hours.

7. Deploy a rapid response team if the patient is in a remote or high-risk location.

Timeliness is critical. The window for effective containment narrows with each day of delay. Contact follow-up should begin immediately, even before laboratory confirmation, and should be stopped if the case is classified as a non case.

3.1.2 Laboratory capacity

Adequate laboratory capacity is fundamental to timely outbreak detection and response. Laboratory confirmation is mandatory for every suspected case. Laboratory confirmation of filovirus disease is essential for activating the full outbreak response and for guiding specific interventions (for example, treatment, vaccination for EVD). Timeliness is therefore critical.

- Specimens should be safely collected and dispatched to the laboratory as soon as a suspected case is identified.

- Laboratory testing should be undertaken immediately, and results should be communicated back to the clinician and public health officer as soon as they are available, to allow for prompt and adapted case management, reporting, investigation and public health response.

In order to achieve that, countries, territories and areas at risk of filovirus disease outbreaks should establish and implement a testing strategy (*4*). Considerations include:

- the designation of a national reference laboratory capacitated to process filovirus specimens;
- testing algorithms enabling filovirus species confirmation and capacity for differential diagnosis;
- clear specimen referral and transportation pathways for peripheral levels with standard operating procedures for sample collection, packaging, storage and transport, cold chain maintenance and chain of custody;
- considerations for decentralization of testing to meet testing needs;
- pre-positioning of specimen collection and transport kits in high-risk areas;
- laboratory reagent stockpiles and cold chain capacity;
- training of personnel involved in all stages of specimen transport on the safe handling and shipment of Category A infectious substances, including appropriate packaging, labelling, documentation and transport procedures; and
- established agreements with regional or international reference laboratories (for example, WHO Collaborating Centres) for confirmatory testing or sequencing (if not available nationally) and for inter-laboratory comparison.

Further guidance on diagnostic testing for filovirus disease can be found in *Diagnostic testing for Ebola disease and Marburg virus disease: interim guidance, 9 July 2026* (*4*).

3.1.3 Wildlife surveillance

Effective animal surveillance, particularly of wildlife and at the human-animal interface, should be a critical component of early warning for filovirus spillover events (*20*) (see Box 4 and Box 5).

Key elements of wildlife surveillance should include:

- syndromic surveillance among wildlife (particularly great apes) and reporting unusual mortality events to veterinary and wildlife authorities;
- active surveillance in high-risk bat species in cave and mine environments where filovirus disease outbreaks have historically occurred;
- sharing of any relevant information among One Health partners, including integration of wildlife disease reporting into national One Health platforms when they exist (see Box 4 and Box 5); and
- collaboration between ministries of health, environment, agriculture and wildlife, and any other relevant institutions.

Box 4. One Health approach to filovirus diseases, key principles

A One Health approach recognizes that human, animal and ecosystem health are deeply interconnected. For filovirus diseases, this means:

- integrating human and animal surveillance to detect early signals of spillover
- monitoring unusual mortality events in wildlife (especially nonhuman primates) as an early warning for human risk
- engaging veterinary, wildlife, forestry and agricultural sectors alongside public health in preparedness and response
- addressing potential drivers of spillover such as land-use change, bushmeat trade and human encroachment into forested areas
- applying risk communication that is grounded in local practices and cultural context.

Box 5. The ZOOSURSY project (*22*)

This project is coordinated by the World Organisation for Animal Health and provides a framework for strengthening surveillance at the animal-human interface in high-risk countries. *"Using a One Health approach, it aims to help strengthen national and regional early detection systems for zoonotic wildlife diseases. Through the better detection, identification and preparedness for future outbreaks of diseases like Ebola and Marburg disease, Rift Valley fever, Crimean-Congo haemorrhagic fever, and Lassa fever, we hope to build a healthier world for animals and humans alike."*

3.2 Surveillance activities during the outbreak

When a filovirus disease outbreak is confirmed, the surveillance system's capacity to detect incident cases must be surged. The structure of the teams involved is described in Annex 1.

3.2.1 Alert system for new incident suspected cases

A multisource, multilevel alert system is essential for rapidly identifying new suspected cases throughout an outbreak, including during mass gatherings. The system should encompass multiple surveillance channels, a defined process for alert validation and case classification, and standardized tools and materials.

Existing surveillance systems should be surged and outbreak-specific mechanisms should be implemented for the duration of the outbreak, using case definitions adapted to the context (suggestions of case definitions during filovirus disease outbreaks are listed in Table 2).

Surveillance systems to surge

- Dedicated event-based surveillance to detect any signal suggestive of filovirus disease.
- Health care facility-based surveillance relying on strengthened epidemiological triage-based standardized case definitions; with clear communication and reporting mechanisms; trained clinical staff; with isolation capacity for suspected cases (see Table 2).
- Strengthened community-based surveillance systems that identify individuals according to the case definitions in Table 2 and/or events associated with transmission (for example, unsafe burials) and report via mobile phone or community leader (*2*). This approach supports the early detection of suspected cases in the community that have not presented to care.
- Points-of-entry (PoE): health authorities at borders, airports and ports screen travellers, particularly from affected areas, for fever and filovirus disease-compatible signs and symptoms.

Additional surveillance systems to implement

- Active case finding run by teams of trained professionals who systematically review health-care facilities' registries and visit communities to actively identify cases.

  All methods for active case finding are detailed in the *Early warning alert and response in emergencies: an operational guide* (section 12.3.4 Step 4: Systematically find cases and contacts) (*2*).
- Contact tracing: the implementation of a contact tracing strategy seeks to detect new suspected cases as soon as possible after symptom onset; contact persons being monitored who develop fever or other symptoms should be immediately classified as suspected cases (see section 4.2.6 Modalities of ending contact person follow-up) (see Table 2).
- Hotline: a national or subnational toll-free hotline should be set up that allows community members, HCWs and others to report suspected cases, unusual deaths and events associated with transmission, such as unsafe burials (see Box 6 and see section 7.3.3 Reporting tools and mechanism for event-based surveillance signals in *Early warning alert and response in emergencies: an operational guide* and Annex 1C of the *Integrated disease surveillance and response technical guidelines, booklet two*) (*2, 17)*.

Table 2. Filovirus case definitions suggested to be used during a filovirus disease outbreak by detecting entity

Filovirus disease case definition to be used for **community-based surveillance** (for example, by a community health worker)	**Signal suggestive of a suspected case (alert)** a. Any individual presenting with fever or recent history of fever and no response to treatment of usual causes of fever in the area; **AND** • at least two of the following symptoms: severe headache, weakness, joint pain, muscle pain, diarrhoea, vomiting, nausea, red eyes/conjunctivitis, hiccups; **OR** • bleeding, including bloody diarrhoea, bleeding from gums, bleeding into skin (purpura), blood in urine. b. Any sudden death (except for deaths following injury or violence). Any such situation should be immediately reported to a surveillance team or to the closest health-care facility.

Table 2. Filovirus case definitions suggested to be used during a filovirus disease outbreak by detecting entity *(cont.)*

Case definitions to be used by HCWs	**Suspected case** a. Any person, alive or dead, with fever or history of fever and no response to usual causes of fever in the area; **AND** • is a filovirus disease case contact person, a dead or sick animal contact person, or a laboratory contact person, as defined in Table 4**; OR** • had travel history in an affected area[2] in the past 21 days **(to be used in countries, territories or areas where there is no active filovirus outbreak).** **OR** b. Any person with fever or history of fever and at least three of the following symptoms: severe headache, weakness, joint pain, muscle pain, diarrhoea, vomiting, nausea, red eyes/conjunctivitis, hiccups, irritability/fussiness (in children under the age of 2). **OR** c. Any person with inexplicable bleeding. **OR** d. Any sudden, inexplicable death. Note: if laboratory specimens are collected in due time during or after the illness, the preceding categories are reclassified as **laboratory confirmed** cases **or non cases**.
	Probable case Any suspected case, alive or deceased, for whom laboratory results were not available, but who has an epidemiological link to a confirmed case or to a recognized transmission chain containing a confirmed case.
	Confirmed case Any suspected or probable case with a positive laboratory result.[3]
	Non case Any suspected or probable case with a negative laboratory result.[3] Negative results include the absence of detectable RNA of, antigens or antibodies to a filovirus, assuming sample was collected in the right time frame.

Fig. 1 illustrates the alert, validation and classification process of cases in filovirus disease surveillance.

As soon as a suspected case is detected, the following steps should be taken.

- Report the case to the surveillance team.
- After obtaining informed consent from the patient or their proxy, trained professionals should collect a sample for laboratory confirmation.
- Fill in a case notification form, including an initial list of contact persons of the suspected case, should the case-patient be subsequently classified as probable or confirmed case.

[2] Affected areas should be defined as precisely as possible at the smallest administrative level to improve the case definition sensitivity.

[3] The preferred assay to confirm a filovirus infection is nucleic acid amplification test; in certain circumstances genomic sequencing or serology (convalescent patients) may be used.

- If **the subject is alive,** explain to the patient and their family the need to go to hospital to receive adequate medical care. After having obtained the consent of the patient or their family, arrange for transfer.

- If **the subject has passed away,** explain to the family the need for conducting a safe burial. After obtaining consent, coordinate funeral arrangements with the burial team.

- At any stage of the algorithm, individuals who evade the response should be actively sought and re-engaged, while respecting the reasons that may have led them to decline investigation, clinical management, contact follow-up, or the safe and dignified burial of deceased relatives. Engagement should be respectful and non-coercive, drawing on the community engagement approaches described in section 6. Community engagement in surveillance activities, with the aim of securing voluntary cooperation with case investigation, clinical care, contact identification and follow-up, and, for deceased relatives, safe and dignified burial.

Case investigation, contact tracing and overall outbreak monitoring will be covered in section 4. Outbreak investigation and contact tracing.

All alerts should be logged and tracked, including those that are invalidated, using the dedicated forms and data collection tools, to enable retrospective analysis of surveillance system performance (see module 14 of *Early warning alert and response in emergencies: an operational guide* (*2*)).

Table 3 lists figures and indicators that should be monitored by the geographical unit and by the source of reporting; they should be interpreted considering other surveillance performance indicators and outbreak investigation findings.

Box 6. Setting up an effective hotline

A dedicated, toll-free hotline is a valuable complement to formal surveillance. Key considerations when implementing a hotline include the following.

- Ensure 24/7 coverage with trained operators following a standardized triage algorithm.

- Establish a rapid response protocol: signals meeting the alert case definition must trigger a field verification visit within 24 hours.

- Track all alerts in a line list: date, location, type of complaint, follow-up action and outcome.

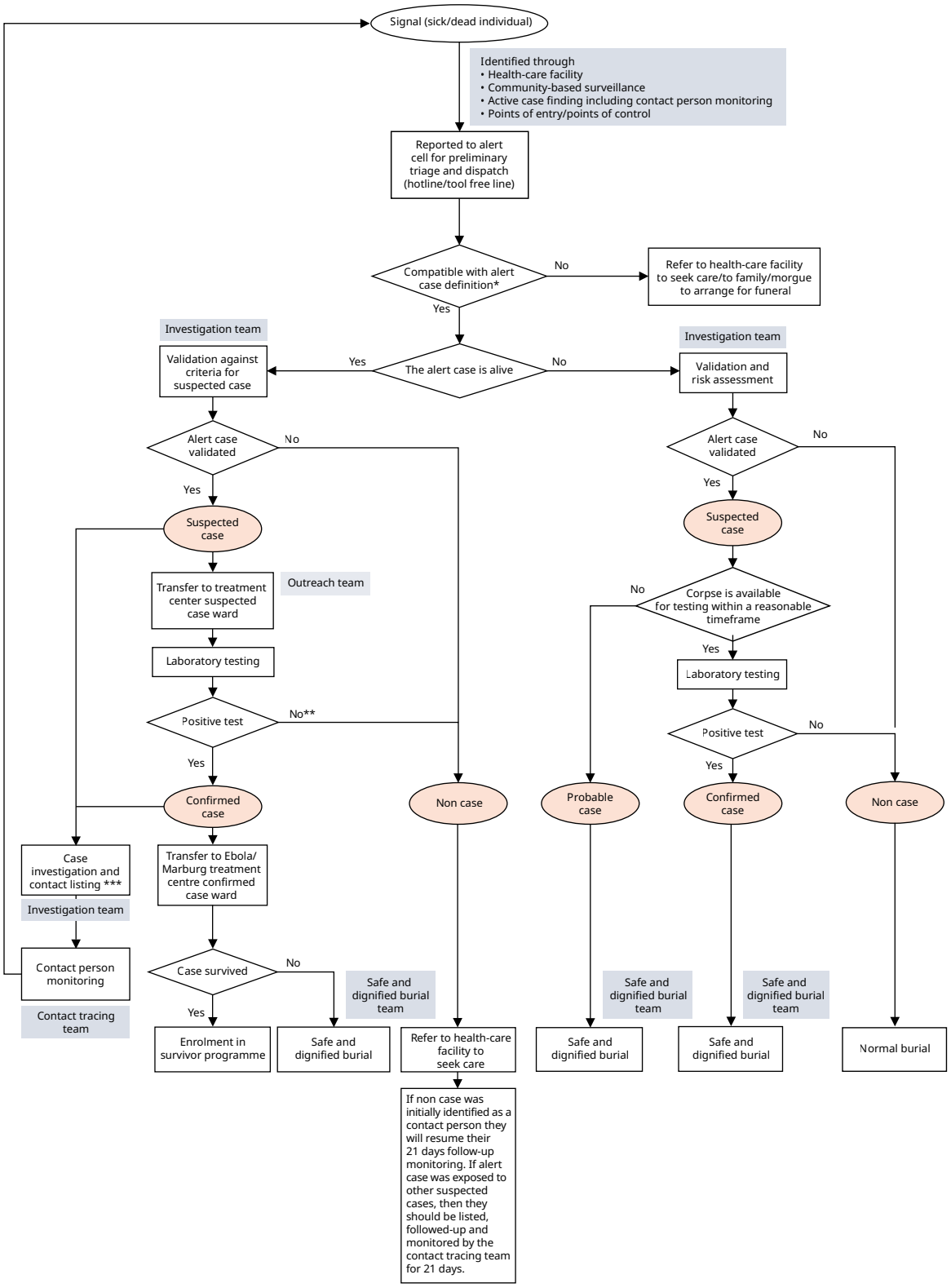
- Promote the hotline number actively through community engagement, radio and local leaders.

- Analyse and use call patterns to identify geographic hotspots or emerging clusters.

- Protect caller anonymity and train operators on respectful, nonstigmatizing communication.

- Consider whether the hotline is intended only to receive information, or whether also to give advice and/or refer callers to emergency/support services. Make sure that the community is well informed about what they can expect when calling the hotline.

Fig. 1. Suggested alert, validation and classification process for filovirus disease cases (to be adapted in different contexts)



* Case definitions to be used during outbreaks are available from Table 2.
** The filovirus disease test (reverse transcription polymerase chain reaction, RT-PCR) may miss the virus in the first three days of symptoms; a negative test before 72 hours doesn't fully rule out filovirus disease; if still sick after 72 hours, the test must be repeated; if recovered within 72 hours, no repeat test is needed (see section 4.2.6).
*** If resources allow, contact listing should be undertaken around suspected cases.

Table 3. Alert surveillance performance figures and indicators to collect and compute during a filovirus disease outbreak (adapted from Table 28, page 154 of *Early warning alert and response in emergencies: an operational guide*) (*2*)

Aggregate numbers	Indicators	Use and interpretation
Number of alerts reported	To be compared to expected number of alerts (in comparison with previous outbreaks)	The total number of reported alerts should be compared to what may be expected. Expected number of alerts may be computed by projecting the number of individuals presenting with similar symptoms based on historical epidemiological data, taking into consideration the number of identified contact persons and what is unknown about the outbreak. The number of alerts will vary over the course of the outbreak.
Number of alerts investigated within 24 hours of reporting	Proportion of alerts investigated within 24 hours (target 100%)	A low proportion would indicate inadequate surveillance capacity (including laboratory) and may need to be escalated. A high indicator may be the sign of good performance; however, it should be interpreted in light of other indicators.
Total number of investigated alerts that were validated as suspected cases	Proportion of investigated alerts that were validated as suspected cases (target 100%)	
Number of suspected cases isolated with adequate IPC precautions	Proportion of suspected cases isolated with adequate IPC precautions (target 100%)	
Number of suspected cases with laboratory samples taken	Proportion of suspected cases with laboratory samples taken with adequate IPC precautions (target 100%)	
	Time from symptom onset to reporting (<24 hours)	Describing the delays by population demographic may help identify bottlenecks.
	Time from symptom onset to isolation (<48 hours)	
	Time from reporting to sample collection (<24 hours)	
	Time from sample collection to laboratory result communication (laboratory turnaround time)	
	Laboratory positivity rate	The number of positive tests out of all the tests performed, broken down by type of specimen (blood, oral swab). This indicator should be interpreted alongside outbreak dynamics and other surveillance performance indicators. For instance, while a low positivity rate may suggest that only a few cases are occurring, it may also be the result of a poor alert system that fails to identify real suspected cases. A high positivity rate may suggest high sensitivity in the application of the suspected case definition.

3.2.2 Concurrent disease monitoring

During a filovirus disease outbreak, surveillance for other priority diseases should be maintained and strengthened. Concurrent outbreaks of vaccine-preventable diseases, including measles, cholera and meningococcal disease, and of malaria, have been documented during filovirus disease responses, often linked to disruption of routine health services and population displacement.

Surveillance teams should do the following.

- Maintain routine reporting of all IDSR priority diseases throughout the outbreak response.
- Track routine vaccination coverage and identify communities at risk of a resurgence of vaccine-preventable diseases.
- Monitor for health-care-seeking behaviour changes that may mask or delay detection of concurrent outbreaks.
- Coordinate with the national expanded programme on immunization on continuity of immunization services.
- Consider mass drug administration campaigns in collaboration with dedicated programmes to limit the number of potential severe malaria cases, which may pose challenges to filovirus disease surveillance, due to the overlapping symptomatology.
- Work with risk communication and community engagement teams for messaging that includes this integrated approach.

4. Outbreak investigation and contact tracing

4.1 Case investigation

Case investigation aims to understand the source of infection of each confirmed and probable case-patient; identify the nature of exposure; and identify other individuals who may have been exposed to the same source or to the case-patient themselves. The case investigation should detect additional cases in the network and enumerate an exhaustive list of contact persons who require follow-up monitoring. All interviews should be prepared and conducted in a culturally sensitive way, ideally in the native language of the case-patient or their proxies.

A standardized case investigation form should be used and completed for every confirmed and probable case as rapidly as possible after case classification; where resources allow, it can also be used on suspected cases while awaiting laboratory results in order to minimize loss of recall. The case investigation form should capture:

- patient demographics: name, age, sex, address, occupation, ethnic group (if appropriate[4]), ongoing or recent pregnancy and outcome (where relevant);
- history of illness: date of symptom onset, signs and symptoms, clinical course, status at time of investigation and/or final outcomes;
- timeline of movements in the 21 days prior to symptom onset (exposure period), including all health-care facilities visited while symptomatic (with dates and treatments received);
- potential exposures: contact with suspected, probable or confirmed cases, sick/dead animals, bats, caves, mines, funeral attendance, health-care facility visits (including traditional healers) during the exposure or illness period (potential nosocomial transmission);
- contact persons of the case-patient during the infectious period (from symptom onset) to be listed using contact person definition criteria;
- vaccination status against filovirus, when available; and
- laboratory testing information and case classification status.

In some situations, the first reported symptoms may not truly be the first symptoms, particularly if the first reported symptoms are those typically associated with advanced disease, such as haemorrhage (see section 5.2.1 Time (*23–25*)). In these situations, out of an abundance of caution, it is worth eliciting contact persons from preceding days rather than strictly from the first reported symptom date.

[4] This will vary by context, and implementation should be guided by national/subnational/local conventions and norms.

In addition, in order to get a complete epidemiological picture of the case-patient, it may be necessary to repeat the interview with the case, including in the treatment centre, or their proxies more than once or to triangulate information from other sources if new information becomes available, such as finding additional case-patients who may have had similar exposures. This is particularly important when conducting in-depth traceback to try to link previously unconnected chains of transmission and rule out more undiscovered transmission chains.

4.2 Contact tracing

Contact tracing is carried out among persons at high risk of infection given likely contact with the case under investigation.

4.2.1 Contact person definition

Effective contact tracing requires the strict application of definitions for what constitutes an exposure to a filovirus disease case in order to promptly identify all and only contact persons (see section 2.4 Transmission: human-to-human). Failure to apply correct definitions can result in missed cases, and thus in ongoing transmission.

This guidance defines contact persons clearly, based on their actual exposure to infected body fluids. There is no category of “high-risk” contact persons with greater exposure than other contact persons. Exposure is binary: an individual either was exposed, in which case they should be informed, monitored and supported throughout the follow-up period, or they were not exposed, in which case no follow-up is needed.

In fact, as per the WHO guideline on contact tracing published in January 2025, a **high-risk contact person** pertains to the consequence an infection may have on the individual themselves. “The high-risk contact person definition is community and context specific and will be determined as a function of the following additional domains: host vulnerability (for example, age, comorbidity, disease outcome if infected, risk factors for severity); and the likelihood of further spread (for example, potential for super-spreader)” (*3*).

Table 4. Filovirus disease contact person definitions suggested to be used (*1, 3*)

Filovirus disease case contact persons	Any person exposed to a probable or confirmed case of filovirus disease within the past 21 days, in at least one of the following ways: • direct physical contact[2] with the case (alive or dead) during the illness or during the funeral, including touching their blood or other body fluids (including breast milk) during the illness or while preparing the corpse; **OR** • indirect contact with broken skin or mucous membranes through surfaces or objects contaminated with body fluids of sick or deceased persons.
Dead or sick animal contact persons	Any person exposed to a sick wild mammal or a carcass within the past 21 days through direct physical contact with the animal (alive or dead) or its blood or body fluids, including carving up the animal or eating raw bush-meat, or reporting a prolonged stay, exposure in mines or caves with bats.
Laboratory contact persons	Any person who has had direct physical contact[5] with specimens collected from confirmed filovirus disease patients or from confirmed filovirus disease animal cases in a laboratory within the past 21 days.

HCWs with a possible occupational exposure to *Orthoebolavirus* or *Orthomarburgvirus* should immediately be assessed for exposure risk using the tool for assessing and categorizing health- and care workers with a possible occupational exposure to *Orthoebolavirus* or *Orthomarburgvirus* in Annex 2 of *Assessment and management of health and care workers with possible occupational exposures to Orthoebolavirus or Orthomarburgvirus: implementation guidance* (*26*). This will determine whether they should be considered as a contact person or not.

4.2.2 Preparing the contact tracing strategy

Before deploying contact tracing teams, a clear operational strategy should be developed, ideally involving community engagement teams who can facilitate feedback from local leaders on acceptable strategies and language to be used. This planning should be done from the outset (see Box 7).

Key considerations include:

- What is the context in which we intend to implement contact tracing? Who are the trusted community members with whom to engage (see Box 8)?
- How will contact persons be notified of their exposure (in person, by phone, through community leaders)?
- What public health and social measures, including movement restrictions, quarantine and monitoring, will be communicated to contact persons, and how will these be explained clearly and respectfully from the start (see Box 9) (*26*)?

[5] Direct physical contact excludes the proper use of personal protective equipment; any damage to personal protective equipment may lead to direct physical contact with infectious material.

- Will contact persons receive any support package (food vouchers, hygiene kits or mobile phones for check-ins)? Note that incentive structures should be determined in advance, applied equitably and backed up with operational planning so that supplies are available and teams can deliver on what is announced.

- How will follow-up monitoring be implemented – in-person daily visits, phone calls, community health worker monitoring or a hybrid approach?

- How will teams handle contact persons in remote areas, across borders or in mobile populations? This includes mass gathering attendees who may be transient, internationally mobile and difficult to trace through conventional means.

- Where will contact tracing data be entered?

Box 7. How contact tracing works

Contact tracing in filovirus disease outbreaks should follow a systematic process:

Step 1: Identify contact persons: using the case investigation form and interviews with the case and their family/community, systematically identify all persons who had exposure to the confirmed or probable case from symptom onset until isolation. You may want to consider drawing family trees to make sure that all contact persons are identified. This information should then be triangulated during interviews with family and neighbours.

Step 2: Notify and register contact persons: each contact person is visited, informed of their potential exposure in a sensitive and non-stigmatizing way and registered with their location and contact details.

Step 3: Daily follow-up for 21 days: each contact person is contacted or visited daily. In-person contact tracing should be encouraged as it creates rapport, permits regular communication and information exchange and helps the direct assessment of the contact person's well-being. Temperature (ideally the contact person would be given a reliable thermometer) and any symptoms are recorded using contact follow-up sheets or software. Any contact person developing fever or symptoms is immediately classified as a new suspected case and isolated. If this contact person tests positive for filovirus infection, they become a confirmed case. If the contact tests negative, they are returned to contact monitoring (see section 4.2.6 Modalities of ending contact person follow-up).

Step 4: Discharge from follow-up monitoring: contact persons who complete 21 days of follow-up are discharged and informed they are no longer at risk from the recent exposure. Their community should also be informed that they no longer represent an infectious risk.

Box 8. Meaningfully engaging with the community

Contact tracing is a strategy that has been used for decades to stop the spread of infectious diseases. Experience has shown that implementing contact tracing successfully requires close and consistent engagement with local communities. This includes having community members (including those selected by the community themselves) as part of contact tracing teams and community involvement in the planning, selection and monitoring of contact tracing methodologies, broader sensitization strategies and communication channels.

Evidence from previous contact tracing efforts shows that across all contexts, community trust is critical for contact tracing to be successful. This trust is even more important for people who may be marginalized or underserved by government or other relevant stakeholder services, particularly in conflict or post-conflict settings or in settings with large refugee or migrant populations.

Strategies and approaches to meaningfully engage with communities to roll out a successful contact tracing strategy are detailed in the *Operational guide for engaging communities in contact tracing, 28 May 2021* (*9*).

Additional reference: Key principles for engaging communities in contact tracing (video).

Box 9. Measures that may be proposed to filovirus disease cases' contact persons

Vaccination (only for EVD)

For outbreaks caused by *Orthoebolavirus zairense*, for which the rVSV-ZEBOV-GP vaccine (Ervebo®) is available, two vaccination strategies may be used.

Ring vaccination strategy

- Ring vaccination targets contact persons of a confirmed or probable case, as well as potential contacts of contact persons, including front-line health-care and response workers as part of the "ring". Potential contact persons are neighbours, family members or extended family members at the closest geographic boundary of all contact persons, plus household members of all contacts.
- Vaccination is offered to all eligible individuals in the ring if previous vaccination was more than 6 months ago.
- Vaccination lists should be drawn from the contact persons' lists to ensure all contact persons have been offered vaccination. Any additional contact persons identified in the course of the vaccination campaign should be fed back as potential contacts to the surveillance team.
- Vaccinated contact persons continue to be followed for 21 days from last exposure (vaccine does not shorten follow-up period).
- Based on the available evidence to date, using the rVSV-ZEBOV-GP, outside of research protocol, is not recommended for the Bundibugyo virus, Sudan virus or MVD outbreaks (*8*).

Box 9. Measures that may be proposed to filovirus disease cases' contact persons *(cont.)*

Targeted geographical vaccination

When ring vaccination is not feasible due to low performance of contact tracing, insecurity or inaccessible areas, a targeted geographical vaccination could be considered as an option or combined with ring vaccination.

- Targeted geographical vaccination is the vaccination of populations living or working in a specific high-risk geographic area around confirmed EVD cases to rapidly reduce local transmission and provide community-level protection.

The WHO International Coordinating Group on vaccine provision manages global vaccine stockpile requests. Early notification of a confirmed EVD outbreak is essential to trigger the stockpile request process.

Considerations for quarantine of contact persons

Quarantine of contact persons (isolation of asymptomatic persons who have been exposed) is a public health and social measure that may be considered for contact persons in specific contexts. It is critical to note that there is no human-to-human transmission prior to symptom onset. Asymptomatic contact persons cannot infect individuals they are in contact with. Therefore, quarantine's unique objective would be limited to referring the suspected case-patient early to clinical management, isolation and testing.

Key principles from WHO public health and social measures guidance include the following.

- Voluntary quarantine with adequate support is more effective and ethically preferable to mandatory quarantine. Securitizing (for example, placing police or soldiers to enforce quarantine) quarantine is rarely successful and significantly decreases trust.
- Quarantine must not result in loss of income, food insecurity or stigma. Social protection measures should be in place: quarantine should be implemented only when there is adequate support in terms of food, water, hygiene, health-care access, psychosocial support, child and elderly care, as well as assurance of continuity of livelihoods (for example, there is someone else available to continue tending fields or caring for livestock).
- Quarantine must be accompanied by clear, respectful communication on purpose, duration and rights. Typically, contact persons should have the right to refuse to quarantine and be given appropriate recommendations in case they do.
- Duration of quarantine corresponds to the incubation period: 21 days from last exposure.
- Quarantine location (home vs designated site) should be determined based on household circumstances and the ability to prevent secondary exposure within the household.
- If institutional quarantine is implemented, it is imperative that contact persons be quarantined in a way that prevents secondary exposure should a contact person develop symptoms (for example, contact persons should not share living, eating or recreational quarters, or toilets).

4.2.3 Human resources for contact tracing

The number of contact tracers required depends on the number of confirmed and probable cases, the average number of contact persons per case, geographic coverage, and the follow-up method used. A general planning ratio is:

- 1 contact tracer per 10–15 contact persons under active follow-up (for in-person daily visits)
- 1 supervisor per 5–8 contact tracers
- additional staff for data entry, logistics and coordination.

These ratios may need to be adjusted to take into account transit times, remoteness of locations, etc.

All contact tracers should receive standardized training covering: the purpose and importance of contact tracing, how to use the contact person listing and monitoring tools, IPC precautions during home visits, communication skills and community engagement, and how to refer a symptomatic contact person for immediate isolation and testing.

Effective supervision of contact tracing teams is critical. Contact tracing supervisors should ensure that protocols are followed consistently and all contact persons under follow-up are adequately monitored. Data quality and timeliness must be ensured. Supervisors should also troubleshoot any operational bottlenecks such as logistics and safety, and community reluctance or refusal to engage with contact tracing.

Contact tracing teams should adopt key health and safety measures as outlined in Box 10.

Box 10. Health and safety precautions for contact tracing teams

- Staff health and safety must be maintained, including access to health care for any medical issues, so that contact tracing teams can effectively perform their roles.
- Although asymptomatic contact persons are not contagious, staff should follow these precautions in case symptoms are hidden or unrecognized.
  - Avoid all physical contact and keep at least a one metre distance at all times.
  - Conduct interviews outdoors. Do not enter a contact person's home unless it is operationally necessary and appropriate IPC precautions can be maintained.
  - Personal protective equipment should not be worn during contact follow-up visits, as it can alarm communities and increase stigma.
- For personal security, staff should do the following.
  - Avoid entering hostile or unwelcoming communities.
  - Never attempt to stop a contact person or case who is trying to flee.
  - Immediately report any problems to their supervisor, who will coordinate support and engage relevant teams (for example, social mobilization) to resolve community issues.

4.2.4 Contact person identification and listing

The following considerations are derived from the 2015 *Implementation and management of contact tracing for Ebola virus disease: emergency guideline* (*1*).

- Laboratory confirmation should not delay the initiation of contact tracing activities.
- Case-patient and contact person interviews should be conducted in a safe and conducive environment to establish trust and rapport between the team, case-patient, family and community (see Box 11).
- The members of the investigation team should be persons trained in contact tracing and interviewing techniques (for example, actively listening, knowing how to ask probing questions, showing empathy and adjusting the interview based on the case-patient's or family members' emotional state).
- When filovirus disease case-patients are first notified of their status, fear can inhibit their ability to recall persons with whom they have had contact. Including psychosocial support staff on the investigation team may help to reassure the case-patient and thereby facilitate contact identification.
- Probing questions should be dependent on the culture and local customs/activities in the area, and they should focus on the period starting from when the case-patient developed symptoms (see section 4.1 Case investigation).
- The contact listing portion of the case investigation form should be designed to systematically elicit the names of all the people with direct physical contact as described in section 2.4 Transmission: human-to-human and Table 4.
- Additional probing questions will aim to identify contexts where direct physical contact may have happened and will need to be further investigated. Therefore, the investigators will seek to obtain the names of:
  - all people who lived or shared meals with the case (alive or deceased) since symptom onset in the same household;
  - all people who visited the case (alive or deceased) since symptom onset (for example, at home or health-care facility);
  - all health-care facilities visited by the case-patient and all HCWs (including traditional healers) who cared for the case-patient;
  - all places the case-patient visited since symptom onset (for example, work, pharmacy, restaurants and bars, shops, places of worship, extended family and traditional healers);
  - if the case-patient is an HCW, the names of all patients and colleagues of this HCW; and, if possible, obtain a duty roster to allow cross-checking of all patients present while the case-patient was working (*20*); and

  - if the case-patient has died, all persons who had contact with the deceased person, including those who prepared the body and/or attended burial ceremonies, including rites that involve direct contact with the body of the deceased.

- It may be helpful to use a calendar of events or specific well-known dates, such as local holidays, to help case-patients or proxies recall activities.

- During initial contact person identification, when relevant, it may be helpful to split the investigation team so that one group focuses on health-care contact persons and one focuses on community contact persons. However, it is important to maintain strong communication between teams and to remember that there may be overlap between community and health-care contact persons.

- Locating all contact persons can be a logistical challenge. Common difficulties include the following: contact persons without addresses, locations with no street names, use of personal nicknames, countries with no national identification programme (for example, no licences or birth certificates) or contact persons that the case-patient does not have information for, such as transportation providers or market vendors. Some solutions to these problems have included engaging community leaders to help find where contact persons live and enlisting the help of cell phone service providers to track contact persons through global positioning systems (subject to data protection safeguards).

Box 11. Informing contact persons of their status

The process of informing the contact person of their status can be traumatizing and should be done with skill, care and empathy.

- Be respectful, honest and direct in your communication. Listen actively and respond in a way that shows support. Answer any questions they may have clearly, using facts.

- Consider the environment where you are having the conversation and, where possible, make sure it is private and quiet and that you have some time to talk.

- Avoid using alarming information, such as “Ebola has no treatment” or “Ebola has a very high case fatality rate”.

- Focus on the benefits of being identified (vs not identified) as a contact, namely, the following.

  - They are more aware of the risks and how to manage them.

  - They have access to immediate medical care if symptoms develop.

    - Symptoms do not necessarily mean they have filovirus disease, but the system allows them access to immediate medical care and prompt filovirus disease testing.

      - If they are positive for filovirus infection, they get immediate access to care and treatment, which can increase the chance of survival.

      - If they are negative for filovirus infection, they will get the testing and medical care they need, as available.

Box 11. Informing contact persons of their status *(cont.)*

- They protect the family, neighbours, community, etc. by avoiding further infections.
- They play a critical role in breaking the chain of transmission in their community/region.
- If contact persons are being quarantined and support such as food is being provided, this should be clearly explained, and it can be a strong support for those who are willing to adhere to contact tracing requirements.

4.2.5 Contact person follow-up monitoring

The contact monitoring team performs contact follow-up through daily visits with the contact person at a predetermined location and time. Contact monitoring teams should ideally be assigned to the same contact person for all 21 days of follow-up. This will ensure continuity and facilitate the development of a relationship, which should, in turn, foster trust. Such an environment will encourage contact persons to report if they develop symptoms. Additionally, this continuity will help identify the details of a contact person's activities of daily living, ensuring that all contact persons are consistently found every day of follow-up.

The following considerations are derived from the 2015 *Implementation and management of contact tracing for Ebola virus disease: emergency guideline* (*1*).

- It is essential that the contact person monitoring team work to build rapport with the contact person in the first few meetings. Trust is essential for contact persons to be truthful regarding their health status and willingness to come forward if they develop symptoms.
- Asymptomatic contact persons are not contagious and can be allowed to continue their activities of daily living during the 21 days of monitoring, provided they can be easily followed up and have timely access to health-care facilities if they become symptomatic.
- Contact persons who are not seen by the monitoring team should be identified and prioritized until they are found, even beyond the 21-day monitoring period.
- A contact person may be unwilling to continue the monitoring process so that they can maintain activities of daily living. This may result in the inability to commit to daily meetings with the contact monitoring team. Contact persons that continue to work in jobs that require interactions with large numbers of people or that impede a contact person's ability to report immediately if symptoms develop (for instance, long-distance bus/truck drivers) should be encouraged to pause work and offered adequate compensation. Measures that can be used to overcome this pressure include the following.
  - Engage community leaders about the importance of the follow-up process and ask for their support.
  - Work with psychosocial workers and social mobilization teams so that they can provide mental and psychosocial support.

  - Explore local solutions to adapt to needs of the community (for example, codesign practical, equitable support that addresses barriers to follow-up without coercion or inappropriate inducement).

  - Provide basic supplies such as food, water, soap and buckets. Note that if supplies are offered, people who are not actually contact persons may come forward.

- Contact persons may flee the area for several reasons. Any lost contact person can undermine the contact tracing process and the ability to stop the ongoing spread of the disease. Contact tracing teams should develop a standard operating procedure for how to classify flight risk in contact persons and document what the correct response should be; and intensify individual, family and community engagement. Understanding the drivers of those wanting to flee the area (for example, stigma, social exclusion and fear), as well as the practical support needs they may have (for example, food, household income and child/family support), can help to proactively mitigate these risks.

- Prevention is the best approach to managing people. This can be done by ensuring contact tracing teams are well trained and can build trusting and respectful relationships with affected communities. People may need practical social protection and livelihood support to adhere to contact tracing requirements.

The following definitions may help identify situations contact tracing teams may face.

A contact person **unreached** is a contact person who has not been reached (physically or by other means) on the day of their follow-up.

A contact person **lost to follow-up** is a contact person who has not been reached for 72 hours or more, after having been followed up for at least one day. All contact persons lost to follow-up should be looked for until they are found, even beyond their 21-day monitoring period. In some instances, human resources will be dedicated specifically to retrieving those contact persons.

A contact person **never-seen** is a contact person who has never been seen.

A **relocated** contact person is a contact person who is now living somewhere other than their original home. This may be within the health district or outside it (for international contact tracing see section 4.2.8 Considerations for international contact tracing).

4.2.6 Modalities of ending contact person follow-up

The contact person develops symptoms within the 21-day follow-up period

When a contact person shows signs of illness during monitoring, the following steps should be taken (see Fig. 1).

1. The contact monitoring team should immediately notify their supervisor, who alerts the field epidemiologist.

2. A team checks if the person shows symptoms compatible with the suspected case definition.

3. If yes, the contact person is taken to an isolation unit for care and testing. Any contact persons of this contact person are listed right away.

4. The new suspected case undergoes testing (see Box 3).

    - In the early stage of the disease (within 72 hours after symptom onset), the RNA of the virus may not yet be detectable. Thus, if the patient remains symptomatic and continues to meet the suspected case definition, the patient must remain isolated and be retested, with collection of a new blood specimen after the initial 72-hour window. If the contact recovers from symptoms within 72 hours, no repeat test is needed.

5. If the test result is

    - negative (after 72 hours post-symptom onset), the person goes home (or back to institutional quarantine) and resumes their original 21-day monitoring as a contact person. If the person was admitted to a filovirus treatment facility, while they were considered a suspected case, and developed filovirus-compatible symptoms afterwards, they must be monitored for 21 days from their discharge date, as they may have been exposed to new cases while in suspected case isolation ward.

    - positive, the person is confirmed as a filovirus disease case-patient and admitted to a treatment unit.

The contact person does not develop symptoms within the 21-day follow-up period

Contact persons are released after being seen and assessed on day 21 with no filovirus disease symptoms. No contact person can be released without a day-21 (or later) evaluation.

Contact persons may also be released if it is verified they were never actually exposed to filovirus disease.

If a contact person is re-exposed to a new filovirus disease case, their 21-day follow-up restarts from the most recent exposure date.

Following release, stigma from families and communities may persist. Health education campaigns should emphasize that released contact persons pose no transmission risk. Official letters confirming end of follow-up should be provided to employers or the community upon request.

4.2.7 Contact tracing performance monitoring

Data on contact person identification should be collected, including demographics, contact details, date of last exposure and information on source case-patient (to link to case line list). Include the type and setting of exposure, daily follow-up and reported symptoms, and outcome of follow-up. The dedicated forms and data collection tools must be used to enable prospective and retrospective analysis of the contact tracing performance. See module 9 and Table 21 of *Early warning alert and response in emergencies: an operational guide* (*2*).

Monitoring the performance of contact tracing is key to driving corrective measures, adapting capacities and investigating barriers. The indicators laid out in Table 5 should be regularly (daily or weekly) computed by the geographic unit and for different demographics, where relevant.

Table 5. Contact tracing performance figures and indicators to compute during a filovirus disease outbreak (*1*, *3*)

Number	Indicators	Use and interpretation
Number of new contact persons identified		This number gives an indication of the level of effort requested from contact tracing teams.
	Mean and median number of contact persons per case-patient	This indicator should be computed by various demographics and can give an idea of how many contacts may be expected for a given case-patient profile or exposure setting. The number of contact persons will depend on the geographic location, culture, living conditions, activity/profession and the amount of time the case was symptomatic before being identified. Be cautious when the numbers do not align with previous cases with similar characteristics.
	Proportion of confirmed and probable cases with contact tracing implemented within 24 hours of suspected case identification (target 100%)	
	Time from case detection to initial contact listing	
Number of contact persons who should be followed up (active contact persons)		This number gives an indication of the level of effort requested from contact tracing teams.
Number of active contact persons effectively reached on a given day	Proportion of contact persons monitored daily (target 100%)	All contact persons should be monitored daily. For each contact person not monitored on a given day, all efforts should be made to find them and collect information on their health status.
Number of active contact persons lost to follow-up		This number gives an indication of ongoing risk of emergence of undetected chains of transmission.
Number of active contact persons lost to follow-up on day 7–14 of follow-up		

Table 5. Contact tracing performance figures and indicators to compute during a filovirus disease outbreak (*1, 3*) *(cont.)*

Number	Indicators	Use and interpretation
	Proportion of contact persons monitored throughout all 21 days (target 100%)	
Number of new cases who were known contact persons at time of detection	Proportion of new cases who were known contact persons at time of detection (target 100%)	A low proportion indicates failure in the capacity to identify contact persons.
Number of new cases who were actively followed up as contact persons at time of detection	Proportion of new cases who were actively followed up as contact persons at time of detection (target 100%)	A low proportion indicates failure to implement a successful contact person follow-up mechanism.
Number of contact tracing teams with reports submitted daily		Any team not submitting their report should be reviewed to understand what the obstacle to timely reporting may be.

4.2.8 Considerations for international contact tracing

In filovirus-affected border regions where communities cross routinely for trade, markets, health-care or seasonal work, international contact tracing is an essential activity. Contact tracing arrangements between neighbouring countries must include joint cross-border surveillance, shared or harmonized case and contact definitions, coordination between local and PoE authorities on both sides and community engagement.

This section sets out considerations for identifying, notifying and following up contact persons whose exposure has an international dimension, whether they arrive from, depart to or move between countries, territories or areas. It also sets out how relevant information should be communicated between health authorities of those countries, territories and areas. The operational principles of contact person identification, notification and follow-up monitoring described above apply equally to international contact persons; what differs is the need for coordination across jurisdictions, utilization of travel and border information where relevant and timely communication between the countries, territories and areas through the National IHR focal point (NFP) network, under the IHR (see Annex 3). Communication between countries, territories and areas on international contact tracing is channelled through the NFP. The following principles should guide international communication regarding cases and contacts.

- **Channel.** Use the NFP-to-NFP route, engaging the relevant WHO IHR contact point, where appropriate, as the authoritative channel for notifying and requesting action on international contact persons. Regional mechanisms, cross-border coordination frameworks and secure information-sharing platforms may support this where they exist.

- **Minimum dataset.** Share a defined minimum dataset for each contact person (identity, date and nature of last exposure, remaining follow-up period, travel itinerary and reliable contact details) that is sufficient for the receiving authority to locate and follow up the person, with the fuller record obtained locally during the follow-up interview.

- **Timeliness.** Transmit information as rapidly as possible; the value of international contact tracing depends on reaching the contact person before, or as soon as possible after, arrival.

- **Data protection.** Handle all personal data in line with the IHR and applicable national and international data-protection law, sharing only what is necessary for public health purposes, and keep it secure and confidential.

- **Ethical, social and human rights considerations.** Information on cases and contact persons should be shared in accordance with applicable legal, ethical and human rights principles. Potential risks of stigma, discrimination, criminalization or other adverse consequences should be considered.

- **Tools and templates.** Use agreed templates and, where available, standardized or digital tools for structured, unambiguous exchange, reducing delays caused by translation or reformatting.

The *Considerations for strengthening international information sharing for tracing and managing infectious disease cases and contact persons: interim guidance* (*19*) describes the four key scenarios adapted for filovirus disease in Table 6.

Table 6. Measures to be taken in countries, territories and areas exporting or importing filovirus disease case-patients or contact persons (adapted from (*18, 19*))

	Departure country, territory or area identifies case or contact person	Arrival country, territory or area identifies case or contact person
A case who travelled internationally The individual is a suspected, probable or confirmed case at time of departure (thus symptomatic)	The **departure country, territory or area** should do the following. • Notify the destination country and any relevant transit countries, territories and areas where public health follow-up may be required as soon as possible through the IHR NFP channel. • Communicate all relevant information on the case-patient and investigation. The **arrival country, territory or area** should initiate case investigation, risk assessment and identification of contacts, including those associated with international conveyances (for example, aircraft, ships, buses or trains), monitoring of identified contacts and investigations to determine the source of transmission (where relevant and in collaboration with the departure country, territory or area).	If the identification happened on board a conveyance, the conveyance operator should notify competent authorities at the PoE before or upon arrival. The **arrival country, territory or area** should do as follows. • At PoE, conduct an initial risk assessment and arrange the safe isolation and referral of the suspected case for laboratory testing and clinical care, applying IPC precautions. • Investigate the source of infection; identify potential contacts and exposure events (applying conveyance-specific exposure assessment, where relevant),[6] including possible exposure in departing and transit countries, territories or areas or onboard conveyances. • Identify and follow up contact persons, according to national guidance. • Share through the NFP network information required to identify contacts in other countries, territories or areas, including the country, territory or area of departure. The **departure country, territory or area** should do as follows. • Support the case investigation, case finding and contact tracing.

[6] Use manifests, seating plans, the Public Health passenger locator form, the advance passenger information and the passenger name record data to identify passengers and crew members that may have been exposed.

Table 6. Measures to be taken in countries, territories and areas exporting or importing filovirus disease case-patients or contact persons (adapted from (*18, 19*)) *(cont.)*

	Departure country, territory or area identifies case or contact person	Arrival country, territory or area identifies case or contact person
A contact person who travelled internationally The individual is an asymptomatic contact person at time of travel	The **departure country, territory or area** should do as follows. • Notify the destination country, territory or area and any relevant transit countries, territories or areas where public health follow-up may be required, as soon as possible through the IHR NFP channel. • Share all relevant contact person follow-up information to the receiving country, territory or area, including identity and reliable contact details, date and nature of last exposure, remaining follow-up period, travel itinerary and dates. • Ensure transfer of follow-up responsibilities without interruption. The **arrival country, territory or area** should take up the follow-up monitoring of the contact person without interruption until day 21 of contact follow-up is completed. They should report the outcome to the departure country.	The **arrival country, territory or area** should do as follows. • Investigate and determine where and how exposure occurred, including through requesting information from the departure country, territory or area. • Notify through NFP network the departure country, territory or area and share relevant information. • Take up follow-up monitoring of the contact person without interruption until 21 days from last exposure is completed. • Monitor for symptom development, relocation or loss to follow-up, and share promptly with relevant authorities. • Report the outcome to the departure country, territory or area upon completion of 21-day monitoring period. The **departure country, territory or area** should do as follows. • Support case finding, source investigation and contact tracing activities in the departure country, territory or area.

The scenarios are based on the following assumptions.

- In the scenarios where the case travels internationally, exposure occurred before arrival in the destination country and the individual showed symptoms before and/or during travel to the arrival country, territory or area.

- In the scenarios where the contact person travels internationally, exposure occurred before and/or during travel to the arrival country, territory or area.

Once identified, an international contact person should be followed for 21 days from last exposure, using the same modalities as for any other contact person (see section 4.2.5 Contact person follow-up monitoring and 4.2.6 Modalities of ending contact person follow-up). The distinctive challenges are maintaining continuity of follow-up when the person travels and making advance arrangements for what happens if the contact person develops symptoms far from the response's usual area of operation.

- **Pre-arranged.** When a contact person under follow-up intends to travel abroad and travel is not restricted, follow-up should be transferred to the destination country before departure. The originating country notifies the destination country with the contact person's details, the date of last exposure, the remaining follow-up period and the travel itinerary so that monitoring continues without interruption on arrival.

- **Contact person who departs without notification.** A contact person may attempt to leave without notifying the public health authorities. To prevent this, the following should be in place: clear, respectful communication about the purpose and duration of follow-up and the risks of interrupting it; adequate support (food, income protection, health-care access) so that follow-up does not impose hardship that drives people to leave; agreed, proportionate movement measures and, where justified, exit screening at PoE consistent with the IHR; and a rapid NFP alert mechanism so that if a contact person does leave, the destination and transit countries are informed immediately.

For routine monitoring, an international contact person is managed like any other. However, the logistics and the subsequent management, should the contact person become a case, require ad hoc arrangements agreed to in advance. These may include activation of diplomatic channels; safe isolation and clinical management in the country, territory or area where the person falls ill; possible medical evacuation under strict IPC conditions; cross-border patient-transfer protocols; and clear financial arrangements determining who bears the costs of isolation, treatment, transport or evacuation. Wherever possible, these arrangements should be discussed between the countries, territories or areas concerned before they are needed.

4.2.9 Considerations for contact tracing strategies in mass gatherings

Mass gatherings held in countries, territories or areas affected by a filovirus disease outbreak, such as international sporting events, religious pilgrimages and large cultural or political events, can amplify transmission and facilitate the international spread of diseases. Contact identification arrangements may be challenging because of large attendee numbers, incomplete registration and rapid onward international travel following the event. Therefore, arrangements to support contact tracing should be established before the event. These include access to attendee registration, ticketing and accommodation records, coordination with event organizers and PoE procedures for communication and follow-up of attendees after the event. Attendees should be informed of symptoms compatible with filovirus outbreaks and advised on actions to be taken should they develop these symptoms within 21 days after the event *(27)*.

5. Analytics

5.1 Data management

Data should be analysed and reported daily or weekly, using a set cut-off date (namely, data as of date X). Before each round of analysis, the data should be checked for updates to missing values (that is, previously unknown outcomes) and cleaned.

Often the data will be spread throughout several databases, such as alert logs, case investigation narrative line lists, filovirus disease treatment facilities, laboratories, contact tracing databases and partner organizations' databases. Ongoing surveillance efforts aim to standardize (see Annex 4) and centralize all data on a unique platform; unfortunately, this is yet to be achieved in all countries, territories and areas.

In addition, occasionally partners may offer their support with more complex analyses. This requires the adoption of a data-sharing agreement that establishes the scope of the use of the data shared.

Ideally all data should be anonymized and kept on secured platforms, with role-specific accesses.

5.2 Simple description of the outbreak

Systematic description of the outbreak in terms of time, place, person and clinical characteristics is the foundation of epidemiological analysis. This descriptive analysis informs hypothesis generation, targeted interventions and communication to decision-makers and communities.

These analyses should be interpreted carefully alongside social science and anthropological intelligence to understand the drivers of transmission in the current outbreak.

5.2.1 Time

- Construct an epidemic curve (epicurve) by date of symptom onset for all confirmed and probable cases, updated daily during the acute phase.
- Calculate key delays: signal-to-alert delay, alert-to-investigation delay, symptom onset-to-laboratory confirmation delay, symptom onset-to-isolation delay.
- Track the reproduction number (Rt) to assess whether the outbreak is growing, stable or declining.
- Document the timing of key interventions relative to the epidemic curve.

When date of symptom onset is not available for all cases, alternatives include date of detection, date of hospitalization, date of confirmation, date of notification or date of death (see section 4.1 Case investigation). There are tools for estimating date of onset from date of wet or haemorrhagic symptoms or date of death (*23–25*). It is critical that the fact that an estimate of the true date of onset is being used in the analyses is disclosed and listed among the limitations.

5.2.2 Place

- Map cases by source of exposure, place of residence, health-care facilities with cases, and places visited during the infectious period. Attributing a case to a country, territory or area may be politically sensitive; rules for attribution should be agreed to ahead of time. Regardless of the place a case is attributed to, it is essential that all places visited 21 days prior to symptom onset, and while they were symptomatic, be documented.
- Track place of death (community, health-care facility, transit or treatment centre).
- Map and categorize the contact persons by location.
- Track population movements, including movements of confirmed cases prior to isolation.
- Identify geographic clustering that may suggest common sources or transmission chains.
- Monitor spread to new geographic areas (districts, regions, countries).
- Evaluate the potential risk of cross-border transmission.

5.2.3 People

- Describe the distribution of cases by case definition (confirmed/probable).
- Describe cases and contact persons by age, sex, gender, community, occupation (with special attention to HCWs, including traditional healers).
- Describe cases by status (alive/dead) at time of detection (for example, proportion of new confirmed and probable cases deceased at time of detection).
- Describe cases by type of exposure.
- Describe chains of transmission.
- Calculate attack rates by group and settings where feasible.
- Track case status: alive, deceased, lost to follow-up, recovered, unknown.
- Determine what proportion of new cases were already registered contact persons at the time of detection (a key indicator of contact tracing completeness).

5.2.4 Clinical characteristics

- Describe the frequency of the main clinical characteristics (particularly relevant to rarer filoviruses).
- Determine whether there needs to be adaptation of case definitions based on most common symptoms (not usually necessary in the midst of an outbreak).
- Evaluate potential risk for further transmission in the community based on available cycle threshold values at time of isolation.
- Compare the outcomes in patients based on various factors, including time from symptom onset to isolation/clinical management, sex, gender, known contact person at time of detection, place of clinical management and treatment received.

5.3 Regular outbreak review

The descriptive analysis should be updated regularly (at least every 48–72 hours during the active outbreak phase) and reviewed in multidisciplinary outbreak management meetings. Key patterns to examine include the following.

- Trends in the epicurve: Is the outbreak accelerating, plateauing or declining? Are there shifts in the affected populations? What are they? Are they real or due to a surveillance artefact?
- Cases and deaths: Categorize by confirmed and probable status, and CFR for total confirmed and probable cases.
- Relative burden of cases: Check relative burden across different geographic locations and specific treatment centres/units and the projection of bed needs.
- Emergence of new transmission chains or geographic spread: This suggests gaps in contact tracing.
- Clustering in specific settings (health-care facilities, households, communities): This indicates specific intervention needs.
- Proportion of case-patients that were known contacts: If proportion is declining, contact tracing coverage may be insufficient.
- Delays in the response cascade: Are laboratory results available within 48 hours of specimen collection? Are contact persons being reached within 24 hours of case identification?
- Risk factors for infection: These include places of exposure and cultural rites that may drive the transmission of the disease. For example, these could include invasive procedures, such as scarification, removal of foetus from a dead pregnant woman's womb, preparation of corpse prior to burials and overall IPC practices in health-care facilities.

Those reviews should be prompted by needs identified by teams working on the ground. Findings should be interpreted by the multidisciplinary group and the recommendation should be clearly identified, actionable, dated and delegated to the relevant operational team. All recommendations and actions, roles and responsibilities should be logged in a shared file to ensure accountability. This approach is detailed in the *Integrated outbreak analytics toolkit* (*6*).

5.4 Contextualized interpretation

Epidemiological data must always be interpreted in the context of local knowledge, community dynamics, cultural practices, mobility patterns, health-care-seeking behaviour, and the history of previous outbreaks. Data analysts should work closely with field teams, community engagement officers, social scientists (for example, anthropologists) and local authorities to ensure that findings are interpreted in a way that reflects reality on the ground.

6. Community engagement in surveillance activities

6.1 Engaging communities in surveillance activities

Community engagement is not a parallel or downstream activity; it is an integral part of every phase of filovirus disease surveillance and response. From the earliest stages of outbreak investigation through to the end of the response, surveillance and epidemiology teams must actively involve communities to ensure understanding, trust and genuine collaboration.

Specific activities include the following:

- sharing findings with communities in accessible, nontechnical language and explaining how these findings are informing the response (*6, 29*);
- consulting communities on the design of contact tracing (*9*), quarantine and other public health and social measures to ensure approaches are culturally acceptable and logistically feasible;
- providing regular feedback on the outbreak situation to community leaders, health committees and community health workers (*6, 29*); and
- actively involving community members in surveillance activities (for example, community-based surveillance reporters and contact tracers from the community).

6.2 Culturally sensitive surveillance activities

Surveillance activities should align with and abide by known and unspoken cultural habits. Therefore, surveillance teams should be thoroughly aware of and sensitive to the context in which they navigate. They should always carry with them communication material in the local language and appropriate formats that explain how surveillance works in their context (for example, explanations of contact tracing and community case definitions).

In addition, when developing the surveillance, case investigation and contact tracing strategy, the surveillance teams should make sure that the communities covered by such activities do not face stigma.

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Annex 1. Surveillance, case investigation and contact tracing teams

Overall surveillance coordination

Surveillance activities must be structured and articulated in a way that enables a fluid data pipeline and smooth communication. Ideally, the surveillance systems during a filovirus outbreak build on the existing architecture. No parallel system should be created. However, surge capacity should be made available for specific tasks.

As mentioned in section 3.2.2 Concurrent disease monitoring, routine monitoring of the IDSR priority diseases should be maintained. Additional resources may need to be mobilized for specific surveillance activities related to the outbreak, such as filovirus alert detection or contact tracing.

The description below pertains to surveillance activities and teams that should be surged during a filovirus outbreak. Available resources referenced in the document describe some of those mechanisms which will therefore not be repeated here.

Roles and responsibilities of all relevant surveillance stakeholders are described in Annex G. Roles and responsibilities of various actors in IDSR in the document *Integrated disease surveillance and response technical guidelines, booklet one: introduction section, third edition* (page 51) (*16*).

Structures or teams involved in the early detection and alert system during a filovirus outbreak (also see section 5. Core functions for early warning, alert and response of *Early warning alert and response in emergencies: an operational guide* (page 33) (*2*))

- Health-care facility: Routine surveillance of communicable diseases in health facilities and laboratories typically forms part of the national surveillance system. As part of their routine work, health-care facilities should report any person responding to the suspected case definition through the alert system (see Table 1). Signals referred by health-care facilities are immediately considered as validated alerts and suspected cases, hence referred to the investigation team for testing, case isolation and management, and for investigation.

- Active case finding, including contact tracing and targeted investigation in health-care facilities (see section 12.3.4.1 Enhance surveillance through passive and active case-finding in *Early warning alert and response in emergencies: an operational guide* (page 134) (*2*))

- Community based surveillance (see Annex E. Guide for establishing community based surveillance and response in *Integrated disease surveillance and response technical guidelines, booklet one: introduction section, third edition* (page 41) (*16*))

- PoE/points of control (see Annex H. Guide for establishing surveillance and response systems at PoE in *Integrated disease surveillance and response technical guidelines, booklet one: introduction section, third edition* (page 58) (*16*))

- Signal and alert management: a dedicated cell may be established to verify the signals, clarify the alerts and organize the dispatch of investigation teams.

Roles and responsibilities of the investigation team

The investigation team is formed of field epidemiologists who are trained in interacting with the community, with sick patients, their families and contact persons. They report back to the lead epidemiologist or any intermediate supervisor. The following tasks fall under their roles and responsibility.

- Interview any potential cases.

- Identify all contacts of a case and interview.

- Assess symptomatic contact persons to determine if they meet the case definition.

- Notify contact persons of their status, explain the follow-up procedure, and provide support.

Additional resources may be found in:

- 2.1 Immediate reportable diseases, conditions and events in *Integrated disease surveillance and response technical guidelines, booklet two: sections 1, 2 and 3, third edition* (*17*)

- 12.2 Forming the investigation team in *Early warning alert and response in emergencies: an operational guide* (*2*)

- The WHO rapid response team competency framework available from https://view.genially.com/63a5d8d22e044800139be8c9.

Structure, roles and responsibilities of the contact tracing team

This overview outlines the core roles that make up an effective Ebola contact tracing response, describing who is needed for each team, their required background and their main responsibilities. It is adapted from Annex 7 in the 2015 WHO/CDC *Implementation and management of contact tracing for Ebola virus disease: emergency guideline* (*1*).

Lead epidemiologist for contact tracing (1 per coordination):

- Role: a senior expert who runs the whole contact tracing effort.

- Responsibilities: manages all field epidemiologists, keeps other response leaders informed, secures resources and confirms case status with the laboratory.

Field epidemiologist (1 per affected district):

- Role: manages the day-to-day tracing work in their district.
- Responsibilities: decides which contact persons need continued follow-up or can be discharged, and coordinates with other teams when a contact person becomes a suspected case.

Data manager (1 per district, plus 1 per coordination if multiple districts involved):

- Role: keeps all contact data accurate and up-to-date.
- Responsibilities: sends daily contact lists to supervisors, spots problems (for example, missed follow-ups, outliers).

Investigation team (minimum 2 people per team):

- Role: mix of epidemiologists, clinicians and psychosocial/communication experts.
- Responsibilities: interviews cases and contact persons, identifies contact persons, informs contact persons of their status and completes the contact listing form.

Supervisor (1 per 5–10 contact follow-up teams, approximately 1–2 villages/neighbourhoods each):

- Role: middle manager between field epidemiologist and contact follow-up teams.
- Responsibilities: assigns teams to contact persons, handles field problems, flags symptomatic contact persons and checks quality of the work.

Contact follow-up team (minimum 2 people per team; team count depends on caseload):

- Role: trusted local community members, ideally speaking the local language.
- Responsibilities: visits each contact person daily for 21 days, checks health status, reports findings and watches for anyone becoming ill.

Other supporting teams will include outreach teams/ambulances that are responsible for transferring suspected cases to isolation units and treatment centres; disinfection teams who will be responsible for cleaning and disinfecting locations and vehicles where cases were; and burial teams who will be responsible for conducting safe and dignified burials of individuals who died of filovirus disease.

Annex 2. Essential tools for surveillance and investigation teams

Data collection tools

- Case and contact person definitions
- Suspected case alert forms (paper and digital)
- Case investigation forms (hard copy and digital)
- Contact listing form (captures identity, location, relationship to case, date and nature of exposure, and risk classification for each contact)
- Contact monitoring form (records daily follow-up results (symptom check, temperature) for each contact over the 21-day period)
- Digital tools: Go.Data (WHO), KoboToolbox, DHIS2, or national equivalents. Ensure tools function offline in areas with poor connectivity

Supporting material

- Maps of the surveillance area (health facility catchment areas, community boundaries)
- Calendar of events
- Secure data transmission tools (encrypted phone/tablet)
- Secure data transmission equipment (encrypted smartphones or tablets with offline-capable apps)

Communication material

- Communication materials explaining the hotline and reporting procedures

Infection prevention and control measures

- Surveillance teams should perform hand hygiene frequently using water and soap or rubbing hands with alcohol hand rub.
- Avoid being in direct physical contact with sick people and keep at least one metre distance.
- Avoid touching environments or surfaces that are contaminated with body fluids.
- Personal protective equipment is not routinely required and should be based on risk assessment.

Annex 3. Relevant IHR provisions for international contact tracing

International contact tracing is underpinned by, and must be conducted in accordance with, the International Health Regulations (2005). Provisions of particular relevance include:

- **Article 4 –** designation of National IHR Focal Points as the standing channel for international communication.
- **Article 23 –** health measures on arrival and departure, including the authority, within the limits set by the Regulations, to require information on a traveller's itinerary and contact details to allow tracing.
- **Articles 30–32 –** treatment of travellers, including public health observation of suspect travellers, the health measures that may be applied on entry, the requirement to treat travellers with respect for their dignity, human rights and fundamental freedoms, and to minimize any discomfort.
- **Article 44 –** collaboration and assistance between States Parties in the detection and assessment of, and response to, events.
- **Article 45 –** treatment of personal data, restricting the processing of health information about identified or identifiable persons to what is necessary and requiring it to be kept confidential and processed fairly.
- **Annex 1 –** core capacities required at designated PoE, including for assessing and managing ill travellers and providing appropriate services.
- **Annex 4 –** obligations of conveyance operators, including facilitating the provision of the passenger and crew information (such as manifests and seating plans) needed for contact tracing.

Annex 4. Digital adaptation kit for surveillance, case investigation and contact tracing in filovirus disease outbreaks

To address the critical pain point of fragmented, inconsistent and unstructured data types collected by disparate response teams during an outbreak, the WHO digital adaptation kit for infectious disease surveillance (volume 1) acts as a unifying framework to sanitize and standardize the information space for emergency response systems. It outlines the generic **personas** (such as contact tracers, clinicians and surveillance officers) and standardized **business processes**, including community-based surveillance (alerts), case registration, laboratory specimen collection, active case search and contact tracing, which operational teams adapt by overlaying filovirus disease-specific requirements (such as the 21-day quarantine follow-up cycle, high-risk exposure history and specific clinical signs of filovirus disease).

By establishing standard data structures, decision-support logic and validation rules, the WHO digital adaptation kit ensures that point-of-care digital tools deployed in the field enforce high data quality and case-tracking consistency while utilizing standard classifications such as the international classification of diseases 11th revision, the international classification of diseases 10th revision, the logical observation identifiers names and codes, or the systematized nomenclature of medicine–clinical terms, to enable real-time **interoperability and data exchange** between outbreak response teams, laboratories and national emergency operation databases.

The filovirus data dictionary is a structured reference tool that standardizes clinical and epidemiological data requirements for digital system implementations, such as district health information software (known as DHIS2) trackers, or open medical record system databases. It serves as the baseline for data model and schema design by defining unique data element identifiers (for example, FILOSRV.C1.DE.5) to represent specific client records and options.

Operational teams should use this dictionary according to their focus areas.

Data managers, epidemiologists and system configuration teams (analytics, quality and database setup)

- **Form and user interface logic configuration**: Use the optionality and explain conditionality columns to build forms. Configure fields marked R (required) as mandatory entry points and implement conditional visibility or display logic for fields marked C (conditional) based on the criteria provided (for example, prompting for vaccination details only if the user answers "Yes" to "Previously vaccinated?").

- **Data quality and logic checks**: Configure front-end entry validations or back-end cleaning routines using constraints from the validation condition column (for example, checking that date ≤ current date or enforcing expected numeric ranges for age or temperature).

- **Interoperability and data exchange**: Leverage the embedded clinical codes (the international classification of diseases, 11th revision, the international classification of diseases, 10th revision, the logical observation identifiers names and codes, the systematized nomenclature of medicine–clinical terms, and the anatomical therapeutic chemical classification) to map data fields to international standards. This mapping is critical for enabling interoperability, automated report building and secure data exchange between local clinical software and national databases (such as DHIS2).

- **Reporting and indicators**: Use the linkages to aggregate indicators column to map data elements directly to public health reporting indicators to automate case counts and case detection metrics.

Clinical and field operations teams (workflow alignment)

- **Intake form alignment**: Map physical case intake, alert and contact tracing forms to follow the logical structure and grouping of the filoviruses and common sheets.

- **Concept standardization**: Reference the description and definition column to resolve any uncertainty on the clinical meaning of a field, ensuring consistent collection of data points (for example, establishing standard definition parameters for fever and specifying axillary vs rectal measurements).

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